

MODEL EXAMINATION

Computer Science Fundamentals

Course: Semester V

Time: 1 Hour

Max. Marks: 40

General Instructions:

1. All questions are compulsory except where an internal choice is provided.
2. Write the question number clearly before each answer.
3. Marks for each question are indicated against it in brackets.
4. Answers to sub-parts of a question may be written separately.

SECTION A — Objective Type (1 mark each)

1. Which data structure follows the First-In-First-Out (FIFO) order?
(a) Stack (b) Queue (c) Tree (d) Graph [1]
2. Expand the abbreviation RAM. [1]
3. State True or False: A primary key column can contain NULL values. [1]
4. Expand the abbreviation HTTP. [1]
5. Which of the following sorting algorithms has $O(n \log n)$ worst-case time complexity?
(a) Bubble Sort (b) Quick Sort (c) Merge Sort (d) Selection Sort [1]

SECTION B — Short Answer (3 marks each)

6. Define time complexity. State why Big-O notation is used. [3]
7. Differentiate between a compiler and an interpreter. (Any three points) [3]
8. What is normalization in a database? Name any two normal forms. [3]
9. Differentiate between TCP and UDP. (Any three points) [3]
10. What is a deadlock? State any two conditions necessary for a deadlock to occur. [3]

SECTION C — Descriptive (5 marks each)

11.

- (a) Explain the working of a hash table. [2]
- (b) What is a collision? Describe any one collision resolution technique. [3]

12. Explain the concept of virtual memory using paging. State any two advantages. [5]

13. What is the OSI reference model? List its seven layers in the correct order. [5]

OR

13. What is the TCP/IP model? Compare it with the OSI reference model. [5]

14.

- (a) Define recursion and give one example. [2]
- (b) Convert the decimal number 45 into its binary equivalent. Show all steps. [3]

*** END OF QUESTION PAPER ***